CHEM 110 Final Exam Study Guide

Built from five past finals (Forms C, D, E and Practice Forms A, B; 145 questions). Contents: 1 Topic map · 2 Constants · 3 Equations · 4 Procedures · 5 Common mistakes · 6 Units & sig figs · 7 Practice set · 8 Answer key (separate section).

1. What the exams test most

Every past final is 100 points of multiple choice: Forms A–D have 30 questions (20 concept questions × 3 pts, then 10 harder/calculation questions × 4 pts); Form E has 25 × 4 pts. A periodic table and data sheet are provided.

Topic	E	D	C	A	B	Total
Thermochemistry & energetics	5	5	7	6	5	**28** (19%)
Biomolecules & drug action	4	4	4	4	4	**20** (14%)
Organic structure, isomers & mechanisms	4	3	4	3	4	**18** (12%)
Atomic structure, electrons & light	3	2	2	5	3	**15** (10%)
Intermolecular forces, phases & solubility	2	2	4	3	2	**13** (9%)
Aqueous reactions & solution stoichiometry	2	3	1	2	4	**12** (8%)
Lewis structures, VSEPR & hybridization	1	3	1	2	3	**10** (7%)
Oxidation states & redox	2	2	2	2	1	**9** (6%)
Ionic bonding, Coulomb's law & lattice energy	1	2	2	1	2	**8** (6%)
Gases	0	2	2	2	1	**7** (5%)
Moles, formulas & composition	1	2	1	0	1	**5** (3%)

Columns: Form E, Form D, Form C, Practice A, Practice B. Each question is counted once under its main skill.

2. Constants

Planck's constant	$h = 6.626 \times 10^{-34}$ J·s
Speed of light	$c = 2.998 \times 10^{8}$ m/s
Avogadro's number	$N_A = 6.022 \times 10^{23}$ mol^{-1}
Gas constant	$R = 0.08206$ L·atm·mol^{-1}·K^{-1} $= 8.314$ J·mol^{-1}·K^{-1}
Pressure	1 atm = 760 torr = 760 mmHg = 101.325 kPa
Temperature	K = °C + 273.15
Specific heat of water (dilute solutions)	$c = 4.18$ J·g^{-1}·K^{-1} (a 1 °C change = a 1 K change)
Length	1 nm = 10^{-9} m; 1 µm = 10^{-6} m; 1 pm = 10^{-12} m

3. Key equations

Photon energy	$E = h\nu = hc/\lambda$; $c = \lambda\nu$ Energy per photon (J). Shorter λ → higher E. For per-mole, multiply by N_A.

Coulomb's law	$E \propto q_1 q_2 / r$ Bigger charges and smaller ions → stronger attraction → larger lattice energy, higher melting point.
Ideal gas law	$PV = nRT$ T in kelvin; with R = 0.08206 use atm and L.
Boyle's law	$P_1V_1 = P_2V_2$ Constant n and T.
Dalton's law	$P_A = X_A P_{total} = n_A RT/V$ Each gas behaves as if alone in the container.
Gas density	$d = P\mathcal{M}/RT$ Heavier molar mass → denser gas at the same P and T.
Henry's law	$S = k_H P_{gas}$ Gas solubility is proportional to its partial pressure above the liquid.
Molarity	M = mol solute / L solution; mmol = mL × M
Mass percent	mass % = (mass of part / mass of whole) × 100
Bond energies	$\Delta H \approx \Sigma$ BDE(bonds broken) − Σ BDE(bonds formed) Breaking bonds costs energy (+); forming bonds releases energy (−).
Enthalpies of formation	$\Delta H°_{rxn} = \Sigma\, n\Delta H°_f$(products) − $\Sigma\, n\Delta H°_f$(reactants) Elements in standard states (O_2(g), Fe(s), ...) have $\Delta H°_f = 0$. Watch the phase: H_2O(l) ≠ H_2O(g).
Calorimetry	$q_{soln} = mc\Delta T$; $\Delta H_{rxn} = -q_{soln}/n$ Temperature drops → endothermic (ΔH > 0). Temperature rises → exothermic (ΔH < 0).
Gibbs free energy	$\Delta G = \Delta H - T\Delta S$ Spontaneous when ΔG < 0.

4. Step-by-step procedures

4.1 Precipitation stoichiometry (limiting reagent)

1. Write the precipitate's formula from the ion charges (e.g., $Mg^{2+} + PO_4^{3-} \rightarrow Mg_3(PO_4)_2$).
2. Write the net ionic equation with coefficients ($3\ Mg^{2+} + 2\ PO_4^{3-} \rightarrow Mg_3(PO_4)_2$).
3. mmol of each ion = mL × M (× ions per formula unit of the salt).
4. Divide each ion's mmol by its coefficient; the smaller result is the limiting ion and equals mmol of precipitate.
5. Grams = mol precipitate × molar mass.

4.2 Concentration of an ion left after precipitation

1. Do steps 1–4 above.
2. mmol left = mmol at start − (coefficient × mmol precipitate).
3. Divide by the TOTAL mixed volume ($V_1 + V_2$), not one solution's volume.
4. Spectator ions (Na^+, K^+, NO_3^-, Cl^- here) are only diluted.

4.3 ΔH from bond dissociation energies

1. Draw full Lewis structures for every reactant and product.
2. List every bond broken (reactants) and formed (products), with counts and multiplicity (C=O vs C–O).
3. Cancel bonds that appear unchanged on both sides (optional, but reduces arithmetic).
4. ΔH = Σ broken − Σ formed; check the sign (most combustions are strongly negative).

4.4 $\Delta H°_{rxn}$ from $\Delta H°_{f}$

1. Balance the equation.
2. Multiply each $\Delta H°_{f}$ by its coefficient (elements = 0).
3. Products minus reactants.
4. For a net ionic reaction use the ions' $\Delta H°_{f}$ values and only the species that actually react.

4.5 Coffee-cup calorimetry

1. Mass of solution = water + dissolved solid (unless told otherwise).
2. $q_{soln} = m\ c\ \Delta T$ with $\Delta T = T_{final} - T_{initial}$ (sign matters).
3. $q_{process} = -q_{soln}$; convert J → kJ.
4. Divide by moles of the substance asked about.

4.6 Mass percent and density → molarity

1. Assume 100 g of solution.
2. g solute = mass %; mol solute = g ÷ molar mass.
3. Volume of solution = 100 g ÷ density (g/mL) → convert mL to L.
4. M = mol ÷ L.

4.7 VSEPR geometry

1. Count valence electrons (add for negative charge, subtract for positive).
2. Place the least electronegative atom in the center; complete octets on outer atoms; leftover electrons go on the central atom (expanded octet allowed for period 3+).
3. Electron domains = bonded atoms + lone pairs on the central atom (a double/triple bond counts once).
4. Electron geometry from domains (2 linear, 3 trigonal planar, 4 tetrahedral, 5 trigonal bipyramidal, 6 octahedral); molecular shape from atoms only.
5. Hybridization: 2 → sp, 3 → sp^2, 4 → sp^3, 5 → sp^3d, 6 → sp^3d^2.

4.8 Assigning oxidation states

1. Free elements = 0; monatomic ions = their charge.
2. F = −1; O = −2 (except peroxides, −1); H = +1 with nonmetals (−1 in metal hydrides).
3. Sum = overall charge; solve for the unknown.
4. Oxidized = number goes up (loses e^-) → it is the reducing agent. Reduced = number goes down → it is the oxidizing agent.

4.9 Net ionic equations

1. Check solubility rules: Group 1, NH_4^+, NO_3^-, CH_3COO^- salts are soluble; most Cl^-/Br^-/I^- soluble except Ag^+, Pb^{2+}, Hg_2^{2+}; most SO_4^{2-} soluble except Ba^{2+}, Pb^{2+}, Sr^{2+}, Ca^{2+} (slightly); most CO_3^{2-}, PO_4^{3-}, OH^-, S^{2-} insoluble unless with Group 1/NH_4^+.
2. Split strong electrolytes (soluble salts, strong acids, strong bases) into ions.
3. Keep solids, liquids (H_2O), gases, and weak electrolytes together.
4. Cancel spectator ions; check atom and charge balance.

4.10 Finding chiral centers

1. Look only at sp^3 carbons (no double bonds, no CH_2 or CH_3).
2. A chiral center has four DIFFERENT groups — compare whole groups, not just the first atom.
3. A molecule with exactly one chiral center is chiral; enantiomers are non-superimposable mirror images.

4.11 Curved-arrow mechanisms

1. Arrows start at electrons (lone pair or bond) and point to where they go.
2. Nucleophile = electron-rich (lone pair, π bond, negative charge); electrophile = electron-poor (δ+ carbon, e.g., C=O carbon).
3. Check formal charges and octets after each step; a carbonyl attack goes through a tetrahedral intermediate.

4.12 Electron configurations of transition-metal ions

1. Write the neutral atom ([Ar] $4s^2$ $3d^n$; Cr and Cu are $4s^1$).
2. Remove 4s electrons FIRST, then 3d.
3. Fill d orbitals by Hund's rule (one in each of the 5 before pairing) and count unpaired electrons.

5. Common mistakes

- Forgetting to convert nm → m ($\times 10^{-9}$) in E = hc/λ, or °C → K in gas laws.
- Flipping the sign in bond-energy problems: it is broken − formed (reactants − products), the reverse of the $\Delta H°_f$ formula.
- Using H_2O(g) data when the equation has H_2O(l) (they differ by 44 kJ/mol).
- Calorimetry sign: a temperature DROP means the process absorbed heat (ΔH > 0).
- Using only one solution's volume after mixing; concentrations after mixing use $V_1 + V_2$.
- Ignoring the precipitate's coefficients when finding the limiting ion (divide mmol by 3 for Mg^{2+} in $Mg_3(PO_4)_2$).
- Removing 3d electrons before 4s when forming transition-metal cations.
- Counting a double bond as two electron domains in VSEPR.
- Thinking breaking bonds releases energy — breaking bonds ALWAYS costs energy; only bond formation releases it.
- Confusing molecular shape with electron geometry (H_2O: tetrahedral electron geometry, bent shape).
- Calling a symmetric molecule with polar bonds (CO_2, CCl_4, BF_3) polar.
- Thinking a CH_2 or a C=C carbon can be a chiral center.
- Mixing up oxidizing and reducing agents: the oxidizing agent is the one REDUCED.
- Rounding too early — keep extra digits until the end, then report the sig figs of the least precise data.

6. Units and significant figures

Energy	J, kJ (1 kJ = 1000 J); per photon in J; per mole in kJ/mol
Pressure	atm, torr, mmHg, kPa
Volume	L, mL (1 L = 1000 mL = 1000 cm^3)
Concentration	M = mol/L; mM = mmol/L (1 M = 1000 mM)
Temperature	K for every gas law and ΔG = ΔH − TΔS; ΔT in °C equals ΔT in K

Specific heat	$J \cdot g^{-1} \cdot K^{-1}$
Enthalpy change	kJ (for the reaction as written) or kJ/mol (per mole of a named substance)

- Multiplying/dividing: keep the fewest significant figures of the data.
- Adding/subtracting: keep the fewest decimal places (this is why $\Delta H°_f$ sums keep one decimal place).
- Exact numbers (coefficients, counted atoms, 1000 mL/L) don't limit sig figs.
- On a multiple-choice exam, sig figs rarely decide the answer — but unit and sign errors often do.

7. Practice set

Original questions in the style of the finals. Multiple-choice: circle one. Calculations: show work and give units. Have a periodic table and calculator. Answers and worked solutions are in Section 8 — try every question first.

Atomic structure, electrons & light

1. [Medium] Which species is isoelectronic with Ar and has the **smallest** radius?

(A) S^{2-}
(B) Cl^{-}
(C) K^{+}
(D) Ca^{2+}

2. [Easy] A photon has a wavelength of 486 nm. What is the energy of **one** photon, in joules?

Answer: _____ J

3. [Easy] Which set of quantum numbers is **not allowed**?

(A) $n = 2, \ell = 1, m_\ell = -1$
(B) $n = 3, \ell = 2, m_\ell = +2$
(C) $n = 2, \ell = 2, m_\ell = 0$
(D) $n = 4, \ell = 0, m_\ell = 0$

4. [Easy] Which photon has the **highest** energy?

(A) Radio, $\lambda = 1$ m
(B) Infrared, $\lambda = 10$ μm
(C) Visible, $\lambda = 500$ nm
(D) Ultraviolet, $\lambda = 200$ nm

5. [Medium] How many unpaired electrons are in the ground-state configuration of Ti^{2+}?

Answer: _____

Lewis structures, VSEPR & hybridization

6. [Medium] What is the molecular geometry of XeF_4?

(A) Tetrahedral
(B) Seesaw
(C) Square planar
(D) Octahedral

7. [Easy] What is the molecular geometry of SF_4?

(A) Tetrahedral
(B) Square planar
(C) Seesaw
(D) Trigonal pyramidal

8. [Medium] Which species has a **trigonal pyramidal** molecular geometry?

(A) BF_3
(B) NO_3^{-}
(C) SO_3^{2-}
(D) CO_3^{2-}

Ionic bonding, Coulomb's law & lattice energy

9. [Easy] Two oppositely charged ions move closer together. What happens?

(A) Potential energy increases; it is endothermic
(B) Potential energy decreases; energy is released
(C) Kinetic energy is converted to potential energy
(D) Nothing changes

10. [Easy] Which compound has the **highest** melting point?

(A) NaCl
(B) MgO
(C) KBr
(D) NaF

11. [Hard] A +2 ion and a −1 ion are moved from distance 2d to distance d. Compared with moving a +1 and −1 pair from 2d to d, the change in potential energy is:

(A) The same
(B) Twice as large and negative
(C) Half as large
(D) Twice as large and positive

Intermolecular forces, phases & solubility

12. [Hard] Heating a saturated solution of a salt whose dissolution is endothermic will most likely:

(A) Make the solution unsaturated (more salt could dissolve)
(B) Cause precipitation
(C) Not change solubility
(D) Make it supersaturated

13. [Medium] When steam at 100 °C condenses to liquid water at 100 °C, the process is:

(A) Endothermic; heat is absorbed
(B) Exothermic; heat is released
(C) Neither; temperature is constant
(D) Endothermic because IMFs break

14. [Medium] In which solvent would NaCl be **most** soluble?

(A) Hexane
(B) CCl_4
(C) Water
(D) Benzene

15. [Medium] Which IMFs can exist between a molecule of benzene (C_6H_6) and a molecule of HCl?

(A) Dispersion only
(B) Dispersion and dipole–induced dipole
(C) Hydrogen bonding
(D) Ion–dipole

Oxidation states & redox

16. [Medium] In 2 $H_2O_2 \rightarrow 2\ H_2O + O_2$, what happens to oxygen?

(A) It is only oxidized
(B) It is only reduced
(C) It is both oxidized and reduced
(D) Its oxidation state does not change

17. [Easy] In which species does sulfur have the **highest** oxidation state?

(A) H_2S
(B) S_8
(C) SO_3^{2-}
(D) SO_4^{2-}

18. [Medium] Which statement about an oxidizing agent is true?

(A) It loses electrons
(B) It is itself reduced
(C) It always contains oxygen
(D) Its oxidation state increases

Aqueous reactions & solution stoichiometry

19. [Medium] Which pair of solutions forms a precipitate when mixed?

(A) $NaNO_3$ + KCl
(B) $AgNO_3$ + NaBr
(C) NH_4Cl + KOH
(D) Na_2SO_4 + KNO_3

20. [Medium] An aqueous solution is 5.00% glucose ($C_6H_{12}O_6$) by mass and has a density of 1.02 g/mL. What is its molarity?

Answer: _____ M

21. [Hard] 40.0 mL of 0.0900 M $NiCl_2$(aq) is mixed with 60.0 mL of 0.0300 M K_3PO_4(aq). Assuming the limiting reagent is completely consumed, how many grams of $Ni_3(PO_4)_2$ precipitate form?

Answer: _____ g

22. [Hard] 40.0 mL of 0.0500 M $FeCl_3$ is mixed with 60.0 mL of 0.200 M NaOH, and $Fe(OH)_3$ precipitates until the limiting ion is used up. What is the final concentration of OH^- (in mM)?

Answer: _____ mM

Moles, formulas & composition

23. [Medium] How many grams of N are in 12.5 g of urea (CH_4N_2O)?

Answer: _____ g

24. [Medium] How many N atoms are in 10.0 g of adenine ($C_5H_5N_5$)?

Answer: _____ atoms

25. [Medium] How many grams of C are in 100.0 g of glucose ($C_6H_{12}O_6$)?

Answer: _____ g

Gases

26. [Easy] A 12.0 L rigid container holds a gas at 0.850 atm and 100.0°C. How many moles of gas are present?

Answer: _____ mol

27. [Medium] A rigid container holds equal moles of N_2 and O_2. More O_2 is added at constant V and T. What happens to the partial pressure of N_2?

(A) Increases
(B) Decreases
(C) Stays the same
(D) Becomes zero

28. [Medium] A 10.0 L flask at 0.0°C contains 1.20 mol CO_2 and 0.300 mol H_2. What is the partial pressure of CO_2?

Answer: _____ atm

Thermochemistry & energetics

29. [Medium] Use the bond dissociation energies to estimate $\Delta H°_{rxn}$ for:

$N_2(g) + 3\ H_2(g) \rightarrow 2\ NH_3(g)$

BDE (kJ/mol) — H–H: 436; N–H: 391; N≡N: 945

Answer: _____ kJ

30. [Medium] Use standard enthalpies of formation to calculate $\Delta H°_{rxn}$ for:

$C_6H_{12}O_6(s) + 6\ O_2(g) \rightarrow 6\ CO_2(g) + 6\ H_2O(l)$

$\Delta H°_f$ (kJ/mol) — $C_6H_{12}O_6(s)$: −1273.3; $CO_2(g)$: −393.5; $H_2O(l)$: −285.8

(Elements in their standard states have $\Delta H°_f = 0$.)

Answer: _____ kJ

31. [Hard] Which statement explains why dissolving glucose in water involves an endothermic step?

(A) Covalent bonds in glucose break
(B) Hydrogen bonds between water molecules must be broken to make room
(C) Ion–ion attractions break
(D) New H-bonds form between glucose and water

32. [Hard] 4.00 g of $KClO_4$ dissolves in water in a coffee-cup calorimeter. The final solution has a total mass of 204.0 g, a specific heat of 4.18 $J·g^{-1}·K^{-1}$, and its temperature drops by 1.42 °C. Calculate ΔH_{soln} in kJ per mole of $KClO_4$.

Answer: _____ kJ/mol

33. [Hard] The combustion of propane (C_3H_8) has $\Delta H = -2220$ kJ per mole of propane. How much heat (in kJ) is released when enough propane burns to produce 50.0 g of CO_2?

Answer: _____ kJ

34. [Easy] Use the bond dissociation energies to estimate $\Delta H°_{rxn}$ for:

$H_2(g) + Br_2(g) \rightarrow 2\ HBr(g)$

BDE (kJ/mol) — Br–Br: 193; H–Br: 366; H–H: 436

Answer: _____ kJ

35. [Easy] Use standard enthalpies of formation to calculate $\Delta H°_{rxn}$ for:

$CaCO_3(s) \rightarrow CaO(s) + CO_2(g)$

$\Delta H°_f$ (kJ/mol) — $CaCO_3(s)$: −1206.9; $CaO(s)$: −635.1; $CO_2(g)$: −393.5

(Elements in their standard states have $\Delta H°_f = 0$.)

Answer: _____ kJ

36. [Medium] Which process absorbs the **most** energy per mole?

(A) $I_2(s) \rightarrow I_2(g)$
(B) $I_2(g) \rightarrow 2\ I(g)$
(C) $2\ I(g) \rightarrow I_2(g)$
(D) $I_2(g) \rightarrow I_2(s)$

37. [Hard] 10.1 g of KNO_3 dissolves in water in a coffee-cup calorimeter. The final solution has a total mass of 260.1 g, a specific heat of 4.18 $J \cdot g^{-1} \cdot K^{-1}$, and its temperature drops by 3.32 °C. Calculate ΔH_{soln} in kJ per mole of KNO_3.
Answer: _____ kJ/mol

Organic structure, isomers & mechanisms

38. [Medium] Which carbon–carbon bond is the **shortest**?

(A) C–C in ethane
(B) C=C in ethene
(C) C≡C in ethyne
(D) They are all equal

39. [Medium] Which is a ketone?

(A) CH_3CH_2CHO
(B) CH_3COCH_3
(C) CH_3COOH
(D) CH_3OCH_3

40. [Medium] How many chiral carbons are in 2-butanol, $CH_3CH(OH)CH_2CH_3$?

(A) 0
(B) 1
(C) 2
(D) 3

41. [Medium] Which statement about resonance structures is correct?

(A) The molecule flips rapidly between them
(B) They differ only in the placement of electrons, not atoms
(C) They are constitutional isomers
(D) They have different molecular formulas

42. [Easy] Which pair are **constitutional (structural) isomers**?

(A) Ethanol (CH_3CH_2OH) and dimethyl ether (CH_3OCH_3)
(B) cis- and trans-2-butene
(C) Ethanol and methanol
(D) Two resonance structures of NO_3^-

43. [Easy] Which functional group is R–C(=O)–O–R′?

(A) Ketone
(B) Ether
(C) Ester
(D) Amide

Biomolecules & drug action

44. [Easy] A peptide is written Gly–Ala–Ser. Which residue is at the N-terminus?

(A) Gly
(B) Ala
(C) Ser
(D) None, peptides have no ends

45. [Medium] Why is proline rarely found in the middle of an α-helix?

(A) Its side chain is charged
(B) The backbone N of proline has no H, so it cannot donate the H-bond the helix needs
(C) It is too large
(D) It forms disulfide bonds

46. [Medium] In a DNA double helix, guanine pairs with:

(A) Adenine, via 2 H-bonds
(B) Cytosine, via 3 H-bonds
(C) Thymine, via 2 H-bonds
(D) Uracil, via 3 H-bonds

47. [Easy] Which amino acid side chain can **hydrogen-bond** with water?

(A) Valine
(B) Leucine
(C) Serine
(D) Phenylalanine

48. [Hard] A drug is redesigned with an extra ring that locks it into its bound shape. Why might binding improve?

(A) The drug gains entropy on binding
(B) The drug loses less conformational entropy when it binds
(C) Rings always make stronger covalent bonds
(D) It decreases the entropy of water

49. [Hard] An enzyme binds L-alanine but not D-alanine. The best explanation is:

(A) D-alanine has a different molecular formula
(B) D-alanine is the mirror image and does not fit the chiral active site
(C) D-alanine has weaker covalent bonds
(D) D-alanine is not an amino acid

8. Answer key with worked solutions

Separate section — fold or detach before working the practice set.

1. Answer: (D) Ca^{2+}

1. All have 18 electrons.
2. Protons: S 16, Cl 17, K 19, Ca 20. Most protons → strongest pull → **Ca^{2+}** smallest.

2. Answer: 4.09×10^{-19} J

1. Convert the wavelength: $\lambda = 486 \text{ nm} \times (1 \times 10^{-9} \text{ m} / 1 \text{ nm}) = 4.86 \times 10^{-7}$ m
2. Use $E = hc/\lambda$ with $h = 6.626 \times 10^{-34}$ J·s and $c = 2.998 \times 10^{8}$ m/s
3. $E = (6.626 \times 10^{-34})(2.998 \times 10^{8}) / (4.86 \times 10^{-7}) =$ **4.09×10^{-19} J**
4. Check: shorter wavelength → higher energy.

3. Answer: (C) $n = 2, \ell = 2, m_\ell = 0$

1. For $n = 2$, ℓ can only be 0 or 1.
2. So $n = 2, \ell = 2$ is **not allowed**.

4. Answer: (D) Ultraviolet, λ = 200 nm

1. Convert to the same unit: 1 m, 1×10^{-5} m, 5×10^{-7} m, 2×10^{-7} m.
2. Shortest λ → **UV, 200 nm**.

5. Answer: 2

1. Ti has Z = 22: [Ar] $4s^2\ 3d^2$
2. Remove 2 electrons, 4s first: Ti^{2+} = [Ar] $3d^2$
3. Fill five d orbitals singly first: 2 electrons → **2 unpaired**

6. Answer: (C) Square planar

1. 4 bonds + 2 lone pairs = 6 electron domains (octahedral electron geometry).
2. Lone pairs sit opposite each other → **square planar**.

7. Answer: (C) Seesaw

1. S has 6 valence e^-; 4 bonds to F use 4, leaving 1 lone pair.
2. 5 electron domains → trigonal bipyramidal electron geometry.
3. Lone pair goes equatorial → molecular geometry is **seesaw**.

8. Answer: (C) SO_3^{2-}

1. BF_3, NO_3^-, CO_3^{2-}: 3 domains, no lone pair → trigonal planar.
2. SO_3^{2-}: S has 3 bonding domains + 1 lone pair → **trigonal pyramidal**.

9. Answer: (B) Potential energy decreases; energy is released

1. With opposite charges, E is negative and becomes MORE negative as d shrinks.
2. Potential energy **decreases** and is converted to kinetic energy/heat → released.

10. Answer: (B) MgO

1. MgO has +2/−2 charges; the others are +1/−1.
2. Charge product 4 vs 1 dominates → **MgO**.

11. Answer: (B) Twice as large and negative

1. For both: $(1/d - 1/2d) = 1/(2d) > 0$.
2. $q_1q_2 = -2$ vs -1, so ΔE is **twice as large and negative**.

12. Answer: (A) Make the solution unsaturated (more salt could dissolve)

1. Solubility increases with temperature for endothermic dissolution.
2. The same amount of dissolved salt is now below the new limit → **unsaturated**.

13. Answer: (B) Exothermic; heat is released

1. Forming IMFs lowers potential energy → energy released.
2. Condensation is **exothermic** (that's why steam burns are severe), even though temperature stays constant.

14. Answer: (C) Water

1. Hexane, CCl_4, and benzene are nonpolar.
2. Water is polar and forms strong ion–dipole interactions → **water**.

15. Answer: (B) Dispersion and dipole–induced dipole

1. Every pair of molecules has dispersion forces.
2. HCl's permanent dipole can induce a dipole in benzene's electron cloud → dipole–induced dipole.
3. Answer: **dispersion and dipole–induced dipole**.

16. Answer: (C) It is both oxidized and reduced

1. In H_2O_2, O = −1.
2. In H_2O, O = −2 (reduced). In O_2, O = 0 (oxidized).
3. Oxygen is **both oxidized and reduced** (a disproportionation).

17. Answer: (D) SO_4^{2-}

1. H_2S: S = −2. S_8: element, S = 0.
2. SO_3^{2-}: S + 3(−2) = −2 → S = +4.
3. SO_4^{2-}: S + 4(−2) = −2 → S = **+6** (most oxidized).

18. Answer: (B) It is itself reduced

1. The oxidizing agent accepts electrons from another species.
2. Gaining electrons = reduction, so the oxidizing agent is **reduced** (its oxidation state decreases).
3. It does not need to contain oxygen (e.g., F_2, Cl_2).

19. Answer: (B) $AgNO_3$ + NaBr

1. $AgNO_3$ + NaBr → AgBr + $NaNO_3$; silver halides (except AgF) are insoluble.
2. **AgBr(s)** precipitates. (NH_4Cl + KOH gives NH_3 + H_2O, but no solid.)

20. Answer: 0.283 M

1. Take 100.0 g of solution: it contains 5.00 g $C_6H_{12}O_6$
2. Moles = 5.00 g ÷ 180.16 g/mol = 0.02775 mol
3. Volume = 100.0 g ÷ 1.02 g/mL = 98.04 mL = 0.09804 L
4. M = 0.02775 mol / 0.09804 L = **0.283 M**

21. Answer: 0.329 g

1. Precipitate: $Ni_3(PO_4)_2$ — net ionic: 3 Ni^{2+}(aq) + 2 PO_4^{3-}(aq) → $Ni_3(PO_4)_2$(s)
2. mmol Ni^{2+} = 40.0 mL × 0.0900 mmol/mL = 3.600 mmol; mmol PO_4^{3-} = 60.0 mL × 0.0300 mmol/mL = 1.800 mmol (each formula unit of salt gives one of these ions)
3. Divide by coefficients: 3.600/3 = 1.200; 1.800/2 = 0.9000 → limiting ion is PO_4^{3-}
4. $Ni_3(PO_4)_2$ formed = 0.9000 mmol = 0.0009000 mol; molar mass = 366.02 g/mol
5. mass = 0.0009000 mol × 366.02 g/mol = **0.329 g**

22. Answer: 60.0 mM

1. Precipitate $Fe(OH)_3$ uses 1 Fe^{3+} : 3 OH^-
2. Start: 2.000 mmol Fe^{3+} and 12.00 mmol OH^-
3. Limiting: divide by coefficients → 2.000/1 = 2.000 vs 12.00/3 = 4.000; the smaller wins, so 2.000 mmol of $Fe(OH)_3$ form
4. OH^- used = 3 × 2.000 = 6.000 mmol; remaining = 12.00 − 6.000 = 6.000 mmol
5. Total volume = 100 mL → $[OH^-]$ = 6.000 mmol / 0.1000 L = **60.0 mM**

23. Answer: 5.83 g

1. Molar mass of CH_4N_2O = 60.06 g/mol
2. Mass of N per mole = 2 × 14.007 = 28.014 g
3. Mass fraction of N = 28.014 / 60.06 = 0.4665
4. Mass of N = 12.5 g × 0.4665 = **5.83 g**

24. Answer: 2.23×10^{23} atoms

1. Molar mass of $C_5H_5N_5$ = 135.13 g/mol
2. Moles of compound = 10.0 g ÷ 135.13 g/mol = 0.07400 mol
3. Each formula unit has 5 N atoms: 0.07400 × 5 = 0.3700 mol N
4. Atoms = 0.3700 mol × 6.022×10^{23} /mol = **2.23×10^{23} atoms**

25. Answer: 40.0 g

1. Molar mass of $C_6H_{12}O_6$ = 180.16 g/mol
2. Mass of C per mole = 6 × 12.011 = 72.066 g
3. Mass fraction of C = 72.066 / 180.16 = 0.4000
4. Mass of C = 100.0 g × 0.4000 = **40.0 g**

26. Answer: 0.333 mol

1. Convert temperature to kelvin: T = 100.0 + 273.15 = 373.15 K
2. n = PV/RT = (0.8500 atm)(12.0 L) / (0.08206 $L \cdot atm \cdot mol^{-1} \cdot K^{-1}$ × 373.15 K)
3. n = **0.333 mol**

27. Answer: (C) Stays the same

1. n(N_2), V, and T are unchanged.
2. So P_{N_2} **stays the same**; its mole fraction drops while total pressure rises.

28. Answer: 2.69 atm

1. T = 273.15 K
2. Mole fraction of CO_2: χ = 1.20/(1.20+0.300) = 0.8000
3. $P_{total} = n_{total}RT/V$ = (1.5)(0.08206)(273.15)/10.0 = 3.362 atm
4. $P_{CO_2} = \chi \times P_{total}$ = **2.69 atm**

29. Answer: −93 kJ

1. Broken (reactants): 1×N≡N (1×945) + 3×H–H (3×436) = 2253 kJ
2. Formed (products): 6×N–H (6×391) = 2346 kJ
3. ΔH ≈ 2253 − 2346 = **−93 kJ**

30. Answer: −2803 kJ

1. Products: 6(−393.5) + 6(−285.8) = −4075.8 kJ
2. Reactants: 1(−1273.3) + 6(0) = −1273.3 kJ
3. ΔH° = (−4075.8) − (−1273.3) = **−2802.5 kJ**

31. Answer: (B) Hydrogen bonds between water molecules must be broken to make room

1. Separating solvent molecules (breaking water–water H-bonds) and solute molecules costs energy.
2. Forming glucose–water H-bonds releases energy (exothermic). Answer: **B**.

32. Answer: 41.9 kJ/mol

1. q_{soln} = m c ΔT = (204.0 g)(4.18 J/g·K)(−1.42 K) = −1211 J
2. Heat for the dissolving process = $-q_{soln}$ = 1211 J = 1.211 kJ
3. mol $KClO_4$ = 4.00 g ÷ 138.54 g/mol = 0.02887 mol
4. ΔH_{soln} = 1.211 kJ / 0.02887 mol = **41.9 kJ/mol** (endothermic: solution got colder)

33. Answer: 841 kJ

1. mol CO_2 = 50.0 g ÷ 44.01 g/mol = 1.1361 mol
2. mol propane = 1.1361 ÷ 3 = 0.37871 mol (1 C_3H_8 → 3 CO_2)
3. heat released = 0.37871 mol × 2220 kJ/mol = **841 kJ**

34. Answer: −103 kJ

1. Broken (reactants): 1×H–H (1×436) + 1×Br–Br (1×193) = 629 kJ
2. Formed (products): 2×H–Br (2×366) = 732 kJ
3. ΔH ≈ 629 − 732 = **−103 kJ**

35. Answer: 178.3 kJ

1. Products: 1(−635.1) + 1(−393.5) = −1028.6 kJ
2. Reactants: 1(−1206.9) = −1206.9 kJ
3. ΔH° = (−1028.6) − (−1206.9) = **178.3 kJ**

36. Answer: (B) $I_2(g) \rightarrow 2\ I(g)$

1. Sublimation overcomes only IMFs (~62 kJ/mol for I_2).
2. Breaking the I–I covalent bond costs ~151 kJ/mol → **$I_2(g) \rightarrow 2\ I(g)$**. Options C and D release energy.

37. Answer: 36.1 kJ/mol

1. q_{soln} = m c ΔT = (260.1 g)(4.18 J/g·K)(−3.32 K) = −3610 J
2. Heat for the dissolving process = $-q_{soln}$ = 3610 J = 3.610 kJ
3. mol KNO_3 = 10.1 g ÷ 101.10 g/mol = 0.09990 mol
4. ΔH_{soln} = 3.610 kJ / 0.09990 mol = **36.1 kJ/mol** (endothermic: solution got colder)

38. Answer: (C) C≡C in ethyne

1. Bond order: 1, 2, 3.
2. Triple bond is **shortest** (~120 pm vs ~134 pm and ~154 pm).

39. Answer: (B) CH_3COCH_3

1. CH_3CHO-type = aldehyde (H on the carbonyl); COOH = acid; C–O–C = ether.
2. $CH_3C(=O)CH_3$ (acetone) → **ketone**.

40. Answer: (B) 1

1. C2 has –H, –OH, $–CH_3$, $–CH_2CH_3$: four different groups → chiral.
2. C1 and C4 (CH_3) and C3 (CH_2, two H's) are not → **1**.

41. Answer: (B) They differ only in the placement of electrons, not atoms

1. Atoms stay in the same positions; only π electrons/lone pairs are placed differently.
2. The real structure is a blend (resonance hybrid) — it does not flip back and forth.

42. Answer: (A) Ethanol (CH_3CH_2OH) and dimethyl ether (CH_3OCH_3)

1. Ethanol and dimethyl ether are both C_2H_6O but the atoms are connected differently → **constitutional isomers**.
2. cis/trans are geometric (stereo)isomers; methanol has a different formula; resonance structures are not different molecules.

43. Answer: (C) Ester

1. Carbonyl carbon bonded to –O–R′ → **ester**.
2. Ketone: C=O between two carbons. Amide: C(=O)–N. Ether: R–O–R′ with no C=O.

44. Answer: (A) Gly

1. The first residue written is the N-terminus → **Gly**; Ser is the C-terminus.

45. Answer: (B) The backbone N of proline has no H, so it cannot donate the H-bond the helix needs

1. Proline's side chain loops back and bonds to its own backbone N, which then carries no H.
2. No N–H → it **cannot donate** a backbone H-bond (and its ring restricts φ).

46. Answer: (B) Cytosine, via 3 H-bonds

1. Base pairs: A–T (2 H-bonds) and **G–C (3 H-bonds)**.

47. Answer: (C) Serine

1. Valine, leucine, and phenylalanine side chains are hydrocarbons (nonpolar).
2. Serine's side chain is $–CH_2OH$ → **H-bonds with water**.

48. Answer: (B) The drug loses less conformational entropy when it binds

1. A pre-organized (rigid) drug already has fewer conformations to lose.
2. So it **loses less entropy on binding** → $\Delta S_{binding}$ less unfavorable → tighter binding.

49. Answer: (B) D-alanine is the mirror image and does not fit the chiral active site

1. L and D are enantiomers: same atoms and bonds, mirror-image 3-D arrangement.
2. A chiral binding site fits only one hand → **shape mismatch**.